CMSC 420: Splay Trees

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1 Problems **2**

1 Problems

1. Show the result of inserting 52 (splay first method) into the splay tree shown below. Show enough detail and provide enough explanation as to clarify your steps.

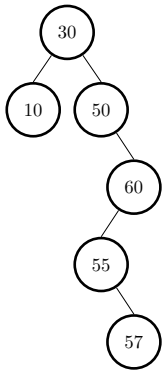

Figure 1: A Splay Tree

2. In a splay tree with height h when we search for a key in a node p, when p is in the tree we know p ends up at the root. It is possible that during this process that some other node gets a free ride closer to the root than it was. What is the maximum number of levels above its original level could some other node end up at? Justify.

3. (a) Explain why there are exactly six possible operations we might implement during each step of a splay process.

   (b) Find the *inverse* of each. In other words for each operation in (a) which operation would undo it?

4. Suppose a node p is inserted into a splay tree (splay first method). We know that p ends up at the root. If r was the original root (before insertion) What is the shallowest and deepest level that r could end up at after insertion of p? Assume the root is level 0. Justify.

5. Suppose x and y are keys/nodes in a splay tree with $x \neq y$. Is it possible that splay(x) could bring y closer to the root than it was? Justify.

6. Suppose a splay tree is undergoing a sequence of operations deletions (splay first method), insertions (splay first method), and searches). The order is unclear except every other operation is a search for a specific key k_0.

   (a) What does this guarantee about the location of k_0? Justify.

   (b) What does this say about the time complexity of searching for k_0? Justify.

7. Show the result of inserting 70 into the splay tree shown below (splay first method). Show enough detail and provide enough explanation as to clarify your steps.

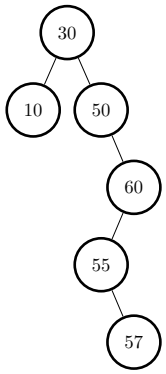

Figure 2: A Splay Tree

8. Show the result of deleting 55 from the splay tree shown below (splay first method). Show enough detail and provide enough explanation as to clarify your steps. For the secondary search use the left subtree.

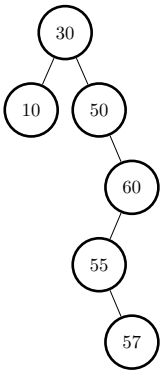

Figure 3: A Splay Tree for Deletion

9. Suppose `x` is the root of a splay tree. Suppose we run this algorithm:

```
while x != None
    while x.left != None:
        splay x.left treating x as the root
        x = x.parent
    end while
    x = x.right
end while
```

What will the splay tree look like after this is run? Explain informally.

10. Show the result of inserting 52 (splay second method) into the splay tree shown below. Show enough detail and provide enough explanation as to clarify your steps.

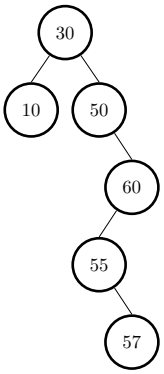

Figure 4: A Splay Tree

11. Show the result of inserting 70 into the splay tree shown below (splay second method). Show enough detail and provide enough explanation as to clarify your steps.

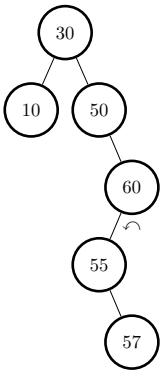

Figure 5: A Splay Tree

12. Show the result of deleting 55 from the splay tree shown below (splay second method). Show enough detail and provide enough explanation as to clarify your steps.

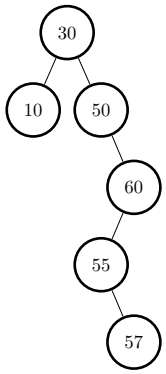

Figure 6: A Splay Tree

13. For each of the following answer in a complete sentence or two. No justification is necessary.

    (a) Suppose we start with a splay tree in which x is the root. We start performing searches. Every third search is for x. What is the lowest level that x could reach?

    (b) Suppose we have a splay tree in which x is the smallest key and y is the second smallest key. We call `splay(x)` followed by `splay(y)`. Where will x and y end up (explain for both)?

    Solution:

14. Suppose you start with a splay tree T containing the keys $1, 2, ..., n$. We know that calling `splay(1)` brings 1 to the root.

    Assume that for some $1 \leq i < n$ if we call `splay(1)`, `splay(2)`, ..., `splay(i)` that the nodes $1, ..., i$ will form a linked list going down and to the left from the root in decreasing order.

    Prove that if we call `splay(1)`, `splay(2)`, ..., `splay(i)`, `splay(i+1)` that the nodes $1, ..., i, i + 1$ will form a linked list going down and to the left from the root in decreasing order.